ACV PROJECT 2026

EmotiSense AI

Real-Time Facial Emotion Recognition System

FastAPI | React + Vite | OpenCV | DeepFace | WebSockets

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Project Overview

What is EmotiSense AI?

EmotiSense AI is an end-to-end Computer Vision pipeline that detects human faces and classifies their emotional state in real-time via a web browser. It uses a client-server architecture with WebSockets for low-latency streaming.

Real-Time Detection

Low-latency WebSocket streaming of webcam frames to backend for instant emotion analysis

Deep Learning AI

VGG-Face model via DeepFace classifies 7 emotions: Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral

Live Dashboard

Cyberpunk UI with Recharts Radar Chart and animated progress bars showing emotion probabilities

System Architecture

End-to-End Computer Vision Pipeline

Browser Webcam
HTML5 Video
+ Canvas API

→

FastAPI Server
WebSocket
Backend

→

OpenCV Detection
Haar Cascade
Face ROI

→

DeepFace VGG-Face
7-Class
Emotion CNN

→

React Frontend
Dashboard
+ Recharts

Pipeline Details

1. Frontend scales frame → JPEG blob → WebSocket binary transmission
2. Server receives frame → cv2.imdecode() → Grayscale conversion → CascadeClassifier face localization
3. Cropped face ROI → DeepFace.analyze(actions=['emotion']) → 7 probability scores via CNN
4. JSON results (bbox + emotions) → WebSocket response → Canvas overlay + Recharts visualization

Technology Stack

Tools, Libraries & Frameworks

⚡ BACKEND

FastAPI / Uvicorn

High-performance async Python web framework

WebSockets

Real-time binary frame streaming protocol

OpenCV-Python

Image manipulation & Haar Cascade face detection

DeepFace + TF-Keras

Pre-trained VGG-Face deep learning model

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WS

🎨 FRONTEND

React + Vite

Fast modern frontend framework with HMR

Tailwind CSS v3

Utility-first dark sci-fi theme styling

Framer Motion

Production-ready animations & transitions

Recharts

Radar Chart & real-time data visualization

Computer Vision Pipeline

Face Detection → Emotion Classification — 7 Emotion Classes

Detection Pipeline

Frame Capture

Frontend scales the webcam frame and converts to JPEG blob — minimizing network bandwidth before WebSocket transmission.

Face Detection

cv2.CascadeClassifier runs on grayscale frame to isolate the facial Region of Interest (ROI) with bounding box coordinates.

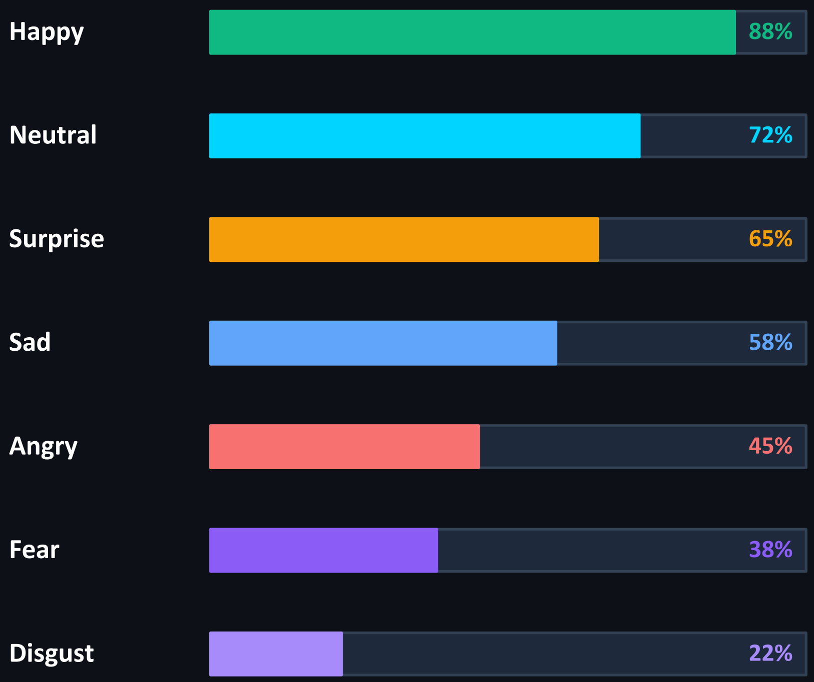
Emotion Classification

Cropped ROI → DeepFace.analyze(actions=['emotion']) → CNN inference → 7 probability scores output.

Overlay & Render

JSON (bbox + emotions) sent back via WebSocket → Canvas overlay draws bounding box + Recharts renders live radar chart.

7 Emotion Classes



**Sample emotion probability distribution*

Setup & Conclusion

Installation Guide & Project Summary

Backend Setup

```
$ cd backend
$ python -m venv venv
$ source venv/bin/activate
$ pip install -r requirements.txt
$ uvicorn main:app --host 0.0.0.0 --port 8000
```

Frontend Setup

```
$ cd frontend
$ npm install
$ npm run dev

→ Open http://localhost:5173
→ Click 'Start Camera'
→ Emotion detection begins!
```

Prerequisites: Python 3.9+ • Node.js 18+ • Webcam • (First run: DeepFace auto-downloads ~6MB model weights)

Project Conclusion

Mayur Chaudhari | Tushar Desale

EmotiSense AI demonstrates a complete real-time AI system integrating Computer Vision, Deep Learning inference, and a modern web interface. It showcases the power of combining OpenCV's classical CV techniques with DeepFace's neural networks for practical emotion-aware applications.